Humanizing *Saccharomyces cerevisiae*: Testing Functional Rescue of Essential Disease Genes for Variant Effect Mapping

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Abstract

The rapid expansion of clinical genome sequencing has revealed extensive genetic variation, yet the functional significance of many variants remains unknown. This gap limits the ability to interpret genetic data in a clinical context, particularly for rare diseases. As a result, there is a growing need for scalable experimental systems that can assess the functional impact of genetic variation and support precision medicine.

Model organisms offer a scalable and robust framework for systematically interrogating human disease genes in an experimentally tractable context. In this study, a humanized *Saccharomyces cerevisiae* platform was used to functionally assess missense mutations in genes associated with rare genetic disease. A three-phase framework was employed: assessment of yeast gene essentiality, validation of functional complementation by the human homolog, and systematic mutagenesis of the human gene of interest to evaluate variant pathogenicity. Of six yeast genes investigated (*SPE3, MPC2, VMS1, PRP4, RLP24,* and *TAM41*), only *TAM41* demonstrated robust complementation with its human homolog, *TAMM41*, enabling downstream mutagenesis and future variant effect mapping. To systematically assess mutation effects, T-REx mutagenesis was employed to generate a library of recombinant constructs, each harboring a single missense mutation within the human gene. These variant libraries were introduced into transgenic diploid yeast, and functional complementation was evaluated following sporulation using selective growth assays. Analysis of 36 missense variants in *TAMM41* revealed a spectrum of functional outcomes, ranging from complete loss-of-function to retained viability. Here, variant effects were assessed using a binary growth phenotype (viable versus non-viable), providing a qualitative measure of function that can be expanded to quantitative metrics in future work.

Together, these findings demonstrate the utility of yeast-based complementation systems coupled with high-throughput mutagenesis for the prospective functional characterization of human genetic variation, while also bringing to light gene-specific limitations in cross-species complementation.

Introduction

The rapid expansion of clinical genome sequencing has revealed hundreds of millions of human genetic variants, yet the functional significance of most remains uncharacterized. This gap is frequently reflected in the classification of variants of uncertain significance (VUS), where insufficient or conflicting evidence prevents clear determination of a variant's role in disease.[1] The challenge is particularly pronounced for rare genetic disorders, many of which are driven by missense mutations that alter protein function but have not been observed in patient populations, limiting opportunities for clinical interpretation and genotype–phenotype correlation.[2] Historically, functional studies have been conducted reactively and on a case-by-case basis, often long after variant discovery. However, recent advances in multiplexed assays now enable the simultaneous assessment of thousands of variants, generating variant effect maps that systematically link genetic variation to functional outcomes. These approaches provide a scalable path toward resolving the disconnect between variant discovery and interpretation, with the long-term goal of building comprehensive resources, such as an Atlas of Variant Effects, to improve disease diagnosis, guide clinical decision-making, and advance precision medicine.[1-4]

Model organisms provide a powerful solution by enabling controlled, high-throughput interrogation of gene function. Among these, *Saccharomyces cerevisiae* is a foundational eukaryotic model due to its genetic tractability, rapid growth, and extensive functional

annotation, with a substantial proportion of yeast genes are evolutionarily conserved in humans, particularly those involved in core cellular processes such as metabolism, RNA processing, and protein homeostasis.[3] This conservation permits yeast to act as a surrogate system for investigating the functional consequences of mutations in human disease-associated genes. However, not all human genes functionally complement their yeast homologs, reflecting evolutionary divergence and differences in cellular context. As a result, establishing functional complementation represents a critical prerequisite for downstream variant analysis, and systematic evaluation of gene compatibility is required to expand the applicability of this platform.

To address this, essential genes, defined by their requirement for cellular viability, can be systematically studied in *S. cerevisiae* using a humanized yeast framework developed by Yellman and colleagues and subsequently expanded by Caron.[3,4] This approach exploits the diploid life cycle of yeast to maintain viability despite deletion of an essential gene, as the deletion is initially confined to a single chromosomal copy. Following sporulation and germination, haploid progeny are generated, in which the essentiality of the functional gene product can be directly assessed.[5] Within this system, functional complementation by a homologous human gene restores viability to haploid yeast lacking the endogenous gene, providing direct experimental evidence of evolutionary and functional conservation. Importantly, the use of essential yeast genes establishes a viability-based assay that enables the interrogation of human genes implicated in rare diseases, forming the foundation for downstream assessment of variant pathogenicity through growth-based phenotypic analysis.[3,4]

To enable systematic interrogation of variant effects, a high-throughput strategy, Tiled-Region Exchange (T-REx) mutagenesis, was developed in our laboratory to generate

comprehensive libraries of missense variants within a human gene of interest.[5] T-REx mutagenesis utilizes degenerate oligonucleotides to introduce mutations across defined tile-regions of a gene, producing a population of constructs in which each clone typically harbors a single amino acid substitution.[5] This approach enables parallel generation of a large number of variants without the need for individual variant construction, which would otherwise be much less efficient. When coupled with a selection-based functional assay, T-REx allows the functional consequences of many mutations to be assessed simultaneously, facilitating the construction of variant effect maps that link sequence variation to protein function.[4,5]

The primary objective of this study is to extend the application of the humanized yeast system to previously untested disease-associated genes, thereby expanding the functional annotation of rare disease variants and contributing predictive insight into mutation severity prior to clinical observation. This study applies this standardized workflow to six yeast genes, selected based on their involvement in conserved cellular pathways and their respective human homologues with known relevance to human diseases (Table 2):

1) **SPE3**, which encodes spermidine synthase, an enzyme responsible for the conversion of putrescine to spermidine in the polyamine biosynthesis pathway; homologous to the human **SRM**;
2) **MPC2**, which encodes a conserved subunit of the mitochondrial pyruvate carrier complex localized to the inner mitochondrial membrane and required for the import of pyruvate into mitochondria, thereby linking cytosolic glycolysis to mitochondrial oxidative metabolism; homologous to the human **MPC2**;
3) **VMS1**, which encodes a stress-responsive protein involved in ribosome-associated quality control and mitochondrial protein homeostasis, functioning in the release and

degradation of stalled polypeptides under conditions of oxidative or mitochondrial stress; homologous to the human **ANKZF1**;

4) **PRP4**, which encodes a serine/threonine protein kinase that is an essential component of the U4/U6–U5 tri-snRNP complex and is required for pre-mRNA splicing and spliceosome activation; homologous to the human **PRPF4**;
5) **RLP24**, which encodes an essential ribosome biogenesis factor associated with pre-60S ribosomal subunits and required for proper large ribosomal subunit maturation and export; homologous to the human **RSL24D1**;
6) **TAM41,** which encodes a mitochondrial phosphatidate cytidylyltransferase localized to the inner mitochondrial membrane and required for cardiolipin biosynthesis, a key phospholipid essential for mitochondrial membrane integrity and oxidative phosphorylation; homologous to the human **TAMM41.**[7,8]

Variable outcomes were observed for functional complementation across the genes tested, reflecting differences in cross-species compatibility. Notably, TAMM41 demonstrated robust complementation, enabling its selection for downstream analysis and variant effect map generation through the assessment of 36 missense variants, including two known pathogenic mutations included for controlling assay validity. While not all genes supported complementation under the conditions tested, their evaluation contributes to defining the scope and limitations of the humanized yeast platform. Collectively, this study establishes a validated framework for the systematic functional characterization of human genetic variation and provides a foundation for the development of a comprehensive data library to expand understanding of the relationship between genotype and function in homologs of essential genes associated with genetic diseases that currently lack effective treatments.

Methods

Saccharomyces cerevisiae is a powerful model organism for human genetic research due to its evolutionary conservation with humans and the relative simplicity of its genome.[6] This study applies a workflow originally developed by Yellman and colleagues, and subsequently established by Caron, extending its application to previously uncharacterized genes.[3,4] The approach exploits the naturally occurring diploid–haploid life cycle of yeast to enable functional analysis of essential genes. Targeted deletions of essential yeast genes are maintained in diploid strains, where viability is preserved by the presence of a second wild-type allele. Upon sporulation and haploid selection, cellular viability becomes dependent on functional complementation.[3,6] Expression of a homologous human gene can rescue haploid yeast lacking the endogenous essential gene, providing direct experimental evidence of conserved gene function. By leveraging yeast ploidy in this manner, this method overcomes the challenge posed by essential gene lethality while enabling systematic investigation of human homologues that were previously inaccessible to functional analysis.[4]

1) Confirming Essentiality

1.1 Selected yeast genes and S. cerevisiae strain

The *S. cerevisiae* strain yAN814 (Table 1) was used to study the essential yeast genes and their corresponding human homologues listed in Table 2. Gene candidates were selected from the Saccharomyces Genome Database (SGD) Literature Archive, drawing from curated lists of yeast genes with reported/predicted human homologues.[7] Several candidates included in this study had previously been reported as non-complementing in yeast-based assays. However, these outcomes are likely attributable to limitations in earlier experimental designs, such pre-existing mutations or deletions which have since been identified.

Table 1. Genotypes of the haploid strains used and their mating counterparts.

Strain ID	Strain Genotype
yAN814	Cross of yAN808 and yAN809 can1::Ste2pr_spHIS5
YAN808	**MATa**, his3d1, ura3d0, leu2d0, lys2d0, RME1pr::ins-308A, lys2d0::LYS2
YAN809	**MATα**, his3d1, ura3d0, leu2d0, lys2d0, RME1pr::ins-308A, can1::Ste2pr_SpHIS5_Ste3pr_LEU2, lys2d0::LYS2

Table 2. Summary of selected yeast genes with their respective human homologues[7,8]

Gene	Human Homolog	Shared Function	Related Human Pathology
SPE3	SRM	Polyamine biosynthesis via spermidine synthase activity	Snyder-Robinson Syndrome (X-linked intellectual disability)[9,10]
MPC2	MPC2	Mitochondrial pyruvate transport and cellular energy metabolism	Mitochondrial Pyruvate Carrier Deficiency (Neurodevelopmental delay, lactic acidosis, hypotonia[11]
VMS1	ANKZF1	Ribosome-associated quality control and mitochondrial proteostasis	Infantile-Onset Inflammatory Bowel Disease[12]
PRP4	PRPF4	Pre-mRNA splicing as a component of the U4/U6–U5 tri-snRNP	Retinitis pigmentosa[13]
RLP24	RSL24D1	Ribosome biogenesis and large subunit (60S) maturation	Diamond-Blackfan Anemia[14]
TAM41	TAMM41	Cardiolipin biosynthesis	Combined oxidative phosphorylation deficiency (COXPD)[15]

1.2 Transforming the Yeast Cells

Homologous recombination was used to generate yeast gene deletions by replacing the target locus with the KanMX cassette, conferring resistance to the antibiotic geneticin (G418). Gene-specific primers flanking the yeast gene of interest were designed to amplify the KanMX cassette from the 2H7a-KlpTEf-Kan-Lox DNA template by polymerase chain reaction (PCR). The upstream primer comprised of 40 bases immediately upstream of the yeast gene start codon fused to 20 bases upstream of the KanMX start codon, while the downstream primer comprised of the reverse complement of 40 bases immediately downstream of the yeast gene stop codon fused to 25 bases downstream of the KanMX stop codon. A standard PCR amplification protocol was performed using SuperFi polymerase (2 U/μL), an annealing temperature of 58 °C, and a 90 second extension time to accommodate amplification of approximately 3,000 bases. The resulting linear KanMX-containing vector was visualized by agarose gel electrophoresis and subsequently transformed into diploid yeast cells following confirmation of successful amplification.

Yeast transformations and subsequent recovery were performed using a rich medium composed of 1% yeast extract, 2% peptone, and 2% dextrose (YPD), with 2% agar added for solid YPD agar plates. A culture of *S. cerevisiae* strain yAN814 (Table 1) was grown to saturation by inoculating a small amount of yeast (from a previously grown on a YPD plate) into 5 mL of liquid YPD medium and incubating overnight at 30 °C on a rolling wheel. To obtain cells in mid-logarithmic phase, 150 μL of the saturated culture was transferred into 5 mL of fresh YPD medium and incubated for an additional 4 hours at 30 °C with rolling.

Cells were harvested by centrifugation, the supernatant was discarded, and the pellet was washed once with 1 mL of sterile water followed by a second centrifugation and removal of the

wash. The transformation mixture was prepared by adding 240 µL of 50% polyethylene glycol 3350 (PEG 3350), 36 µL of 1 M lithium acetate (LiOAc), 50 µL of salmon sperm DNA (previously thawed from −20 °C, denatured by boiling at 95 °C for 5 minutes, and immediately placed on ice), and 20 µL of the KanMX-containing PCR product directly to the cell pellet. The mixture was vortexed vigorously to resuspend the cells and incubated at 42 °C for 55 minutes to induce heat shock.

Following heat shock, cells were pelleted by centrifugation, the supernatant was removed, and the pellet was resuspended in 200 µL of sterile water. The entire suspension was plated onto a YPD agar plate using sterile glass beads to ensure even distribution across the surface with gentle shaking. Plates were then incubated at 30 °C for approximately 30 hours for adequate lawn growth. Transformants were then replica plated by transferring cells onto a velvet cloth and stamping onto a fresh YPD agar supplemented with 0.02% G418. Plates were incubated at 30 °C, and after two days, G418-resistant colonies were observed.

Individual colonies were purified by streaking onto fresh YPD+G418 plates. A single purified colony was subsequently grown in liquid YPD medium at 30 °C for 24 hours. For long-term storage, 900 µL of the saturated diploid culture carrying a heterozygous gene deletion was mixed with 100 µL of cryoprotectant (80% glycerol) and stored at −80 °C.

1.3 Sporulating the Transformed Cells

100 µL of the saturated diploid culture carrying a heterozygous gene deletion was pelleted, then resuspended into 1 mL of sporulation medium, SPO++, consisting of 5% (v/v) of 5× yeast extract, 0.625% (v/v) of 40% dextrose, 20% (v/v) of 5× potassium acetate (KoAc), 1% (v/v) of 100× zinc acetate, 2% (v/v) of 50× uracil supplement, and 1% (v/v) of 100× amino acid supplement containing histidine, lysine, and leucine, with sterile water added to 100% final

volume. This was transferred to a glass tube placed on a rolling wheel at room temperature for 4-5 days.

Germination plates were prepared to selectively recover haploid, plasmid-transformed yeast following sporulation. Synthetic defined medium lacking histidine and supplemented with canavanine (SD-his+can) was used, where canavanine, an arginine analog toxic to diploid cells, selectively eliminates diploids for haploid progeny. The absence of histidine further selects for MATα haploids, as MATa cells in this strain background are histidine auxotrophs, thereby preventing mating and diploid reformation. SD-his+can plates were prepared by supplementing melted agar with 100 mL of 5× yeast nitrogen base with ammonium sulfate (YNB+AS), 20 mL of 25× amino acid mix, 25 mL of 20× D-glucose, 5 mL each of 100× lysine, leucine, methionine, and tryptophan, 10 mL of 50× uracil, and 500 µL of canavanine, with sterile water added to final volume. SMSG-D-his+can+G418 plates were prepared similarly, except that monosodium glutamate (MSG) was used in place of ammonium sulfate as the nitrogen source, and G418 was included for selection.

Sporulated cultures were smeared onto SD-his+can germination plates and incubated at 30 °C for 48 hours to allow haploid colony formation. The germination plate for each gene was then stamped on a velvet cloth and replica plated sequentially onto YPD, then SMSG-D-his+can+G418, being mindful of the plate orientation. Plates were incubated at 30ºC for 24 hours to allow adequate colony growth. The growth of the colonies on the germination selection plate was then recorded as an indicator of essentiality, with growth on YPD serving as positive control for cell viability. Figure 1 illustrates this experimental workflow. For genes identified as non-essential under standard growth conditions, alternative conditions as outlined in

Table 3 were established to induce conditional essentiality, under which functional rescue with the human gene was subsequently assessed.

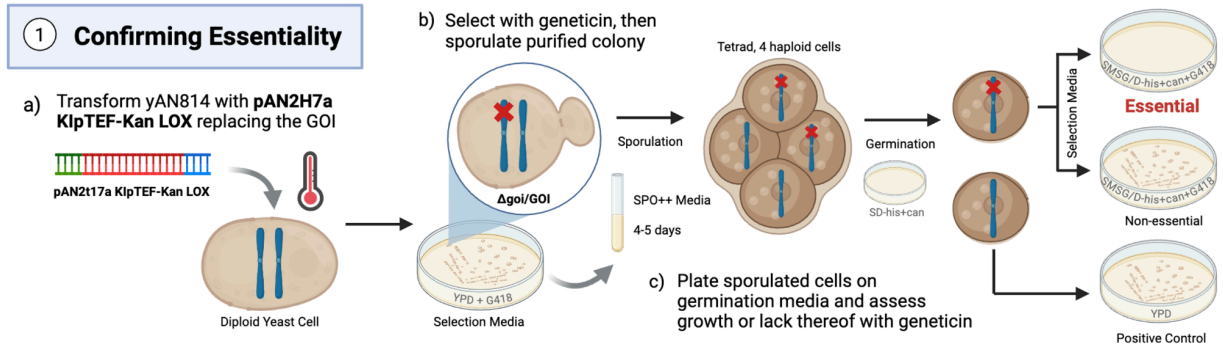


Figure 1. *Schematic workflow for the deletion of an S. cerevisiae gene of interest in preparation for replacement with its respective human homolog.*[4] *Created with BioRender.*

Table 3. *Summary of selected yeast genes with conditions conferring essentiality.*[7,8]

Gene	Essential Conditions	Functional Rationale
SPE3	Absence of panethenoic acid	Involved in polyamine biosynthesis; becomes essential when cofactor availability is limited
MPC2	Absence of leucine	Functions in mitochondrial pyruvate transport; required under conditions dependent on mitochondrial metabolism
VMS1	Heat stress at 37°C	Involved in mitochondrial protein quality control; required under proteotoxic stress
PRP4	Standard	Required for pre-mRNA splicing; essential for gene expression
RLP24	Standard	Involved in ribosome biogenesis; essential for protein synthesis
TAM41	Standard	Required for cardiolipin biosynthesis and mitochondrial function

2) Verifying the Human Homologous Gene Rescues Essential Yeast Gene Function

2.1 Creating the Plasmid Vectors for the Yeast Gene with Promoter and Terminator Regions

Following confirmation of gene essentiality, humanized yeast plasmids were assembled for functional complementation assays. Plasmid vectors containing each yeast gene of interest with its native regulatory elements were constructed for downstream cloning and transformation. Promoter regions were identified as approximately 500 bases upstream of the yeast open reading frame (ORF), and terminating at the nearest upstream start or stop codon of an adjacent gene. Terminator regions were defined as approximately 200 bases downstream of the stop codon, ensuring no overlap with the downstream gene ORF. This strategy was based on the assumption that the ORFs of immediately flanking genes do not contain the promoter or terminator elements, such that regulatory elements reside within the intergenic regions flanking the ORF.

Primers were designed to amplify the full yeast gene including its promoter and terminator regions for cloning into *E. coli*. Forward primers annealed at the beginning of the defined promoter region, while reverse primers annealed at the end of the defined terminator region. Primers incorporated the last 20 bases of SP6 fused to the first 18-30 bases of the promoter region and the last 18-30 bases of the reverse transcript of the terminator region fused with the first 20 bases of T7, Primer GC content was maintained as close to 50% as possible, with melting temperatures approaching 60 °C to ensure efficient and specific amplification. Standard PCR was performed using BY4741 yeast genomic DNA template and amplification was confirmed by gel electrophoresis.

The self replicating plasmid pAN316a was amplified by PCR using SP6 and T7 primers to generate a linear plasmid backbone, also confirmed with gel electrophoresis. Following

amplification, 0.5 µL of DpnI was added to both the gene and plasmid PCR products and incubated at room temperature overnight to digest methylated template DNA.

PCR products were purified using magnetic bead-based purification (NaCl/PEG8000). 50 µL of suspended beads(for approximate 1:1 bead-to-sample ratio) were added to each PCR reaction and mixed thoroughly. Samples were incubated on a magnetic rack for 5 min to allow DNA binding, and the supernatant was removed. Beads were washed with 180 µL of 85% ethanol and air-dried. DNA was eluted in 50 µL of elution buffer, and the purified PCR product was recovered following magnetic separation.

Purified gene and plasmid backbone PCR products were assembled together into a plasmid via Gibson Assembly using 2.5 µL of each purified PCR fragment and 5 µL of 2x Gibson mix. Assembly reactions were incubated in a thermocycler, and the resulting plasmids were subsequently used for transformation into *E. coli.*

2.2 Vector Transfection and Amplification in E. coli

Following Gibson assembly, recombinant plasmids were transformed into chemically competent *E. coli* DH5α cells using a standard heat-shock transformation protocol. Competent cells were thawed on ice, and 1 µL of the assembled plasmid was added directly to the cells and mixed gently by flicking. Cells were incubated on ice for 15–30 minutes, heat shocked at 42 °C for 30 seconds, and immediately returned to ice for 5–10 minutes. Recovery was initiated by adding 500 µL of SOC medium and incubating at 37 °C for 1 hour to allow membrane recovery and expression of ampicillin resistance. SOC recovery medium consisted of Super Optimal Broth (SOB; 2.0% tryptone, 0.5% yeast extract, 10 mM NaCl, 2.5 mM KCl, 10 mM $MgCl_2$, and 10 mM $MgSO_4$) supplemented with 0.36% (w/v) glucose.

Following recovery, 50 µL of the transformation culture diluted with 50 µL sterile water was plated onto LB–Amp agar plates (1.0% tryptone, 0.5% yeast extract, 1.0% NaCl, 1.5% agar, and 100 µg/mL ampicillin) using glass beads for even spreading. Plates were incubated at 37 °C for 24 hours, after which many individual colonies were observed, indicating successful plasmid uptake. Colonies corresponding to each gene construct were then selected and grown in 5 mL of standard ZYM505 medium at 37 ºC for 24 hours in a rolling wheel.[15] Plasmid DNA was extracted using the Geneaid Presto Mini Plasmid Kit and confirmed using gel electrophoresis.[17] Further verification was performed by restriction enzyme digestion using a standard protocol containing 14 µL of sterile water, 6 µL of plasmid DNA, 2 µL of 10x rCutsmart buffer, and 0.5 µL of the selected restriction enzyme, followed by incubation at 37º C for 1 hour and analysis gel electrophoresis. Restriction enzymes used for each construct are listed in Figure 5. Verified plasmids were preserved by mixing 800 µL of the *E. coli* culture with 200 µL of a cryoprotectant solution containing 50% glycerol and 35% DMSO and stored at −80 °C (Figure 2a).

2.3 Optimizing Human Homologues

Human homologues of each yeast gene were identified using UniProt and independently confirmed using the Saccharomyces Genome Database (SGD).[6,7] Homologue selection prioritized conservation of essential functional domains between human and yeast proteins. When multiple human isoforms were present, the isoform with the greatest sequence and domain similarity to the yeast protein was chosen to maximize the likelihood of functional complementation. The selected sequences were reverse-translated and codon-optimized for yeast and ordered.

2.4 Inserting the Human Gene into the Empty Vector ORF

Primers were designed to amplify the plasmid backbone of each gene, including the native yeast promoter and terminator regions while excluding the ORF (Figure 2b). The reverse promoter primer corresponded to the reverse complement of the final 20–35 bp of the promoter region, and the forward terminator primer corresponded to the first 20–35 bp of the terminator region. Primer lengths were selected again to achieve 50% GC content and melting temperatures as close as possible to 60 °C. These primers were used with the corresponding extracted plasmids as templates to amplify the vector lacking the ORF using standard PCR, and confirmed by gel electrophoresis.

PCR products were treated with 0.5 µL DpnI overnight at room temperature, followed by bead purification. The purified vector was then assembled with the codon-optimized human gene by Gibson assembly (Figure 3b) and transformed into *E. coli* DH5α cells using the standard transformation protocol as previously outlined. Transformants were plated on LB–Amp plates and incubated at 37 °C for 24 hours (Figure 2c).

For each gene, three colonies were then selected and cultured individually in 5 mL ZYM505 medium at 37 °C for 18–24 hours with rotation. Plasmids were extracted using the Geneaid Presto Mini Plasmid Kit, and restriction enzyme digestion was performed as previously described using enzymes selected to generate diagnostic fragment patterns confirming insertion of the human gene of interest. Digestion products were visualized by agarose gel electrophoresis.

To ensure sequence fidelity, all humanized plasmids were further validated by Oxford Nanopore PromethION sequencing prior to transformation into yeast for functional complementation assays. 7 µL sequencing samples containing 2 µL of plasmid DNA were prepared. Each plasmid was sequenced in duplicate using SP6 and T7 primers to improve sequencing accuracy.

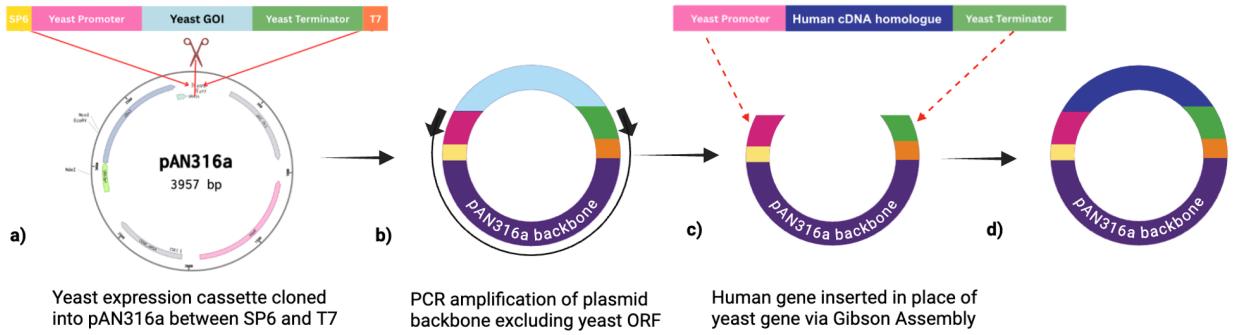


Figure 2. *Schematic workflow for constructing a humanized expression plasmid using PCR and Gibson Assembly.* ***a)*** *Insertion of a yeast gene flanked by its native promoter and terminator into pAN316A between the SP6 and T7 sites, generating a yeast expression cassette;* ***b)*** *The plasmid backbone was PCR-amplified to produce a linear fragment excluding the yeast open reading frame (ORF), while retaining the promoter and terminator regions;* ***c)*** *A human gene of interest, designed with homologous overlaps to the promoter and terminator sequences, was inserted into the linearized backbone via Gibson assembly, resulting in a recombinant plasmid in which the human gene is expressed under yeast regulatory elements;* ***d)*** *Humanized yeast plasmid.*

2.5 Assessing Functional Rescue with Human Gene Complementation

To validate the suitability of the yeast-based platform, the humanized plasmid (pAN316a vector with the human gene flanked by the yeast promoter and terminator regions, Section 2.4) was transformed into its corresponding heterozygous yeast deletion strain (Section 1.2), in which one chromosomal copy of the essential gene is replaced by a KanMX cassette. Diploid yeast cells stored at -80 °C were revived in 5 mL YPD+G418 and grown for 24 hours prior to performing the standard yeast transformation protocol.

Transformants were plated on SD-uracil plates to select for cells containing the plasmid, then incubated at 30 °C for 48 hours. Individual colonies were inoculated into 5 mL liquid

SMSG-D-ura+G418 and grown at 30 °C with rotation for 24 hours, after which sporulation was induced using standard conditions from a saturated culture. Following 4–5 days, spores were plated onto SD-his+can-ura germination plates and incubated at 30 °C for 48 hours. Colonies were subsequently replica plated onto YPD and selective SMSG-D-his+can-ura+G418 plates.

In parallel, transformations were performed using a plasmid expressing the native yeast gene (Figure 2b) to confirm that viability could be restored under the assay conditions, and pAN316a lacking an insert (Figure 2a) to verify that loss of the gene is lethal in the absence of functional complementation. Restoration of growth after 24 hours on selective media following sporulation indicated that the human gene can sufficiently substitute for the yeast homolog, supporting functional conservation and validating the use of this system for downstream analysis of human gene variants (refer to Figure 3).

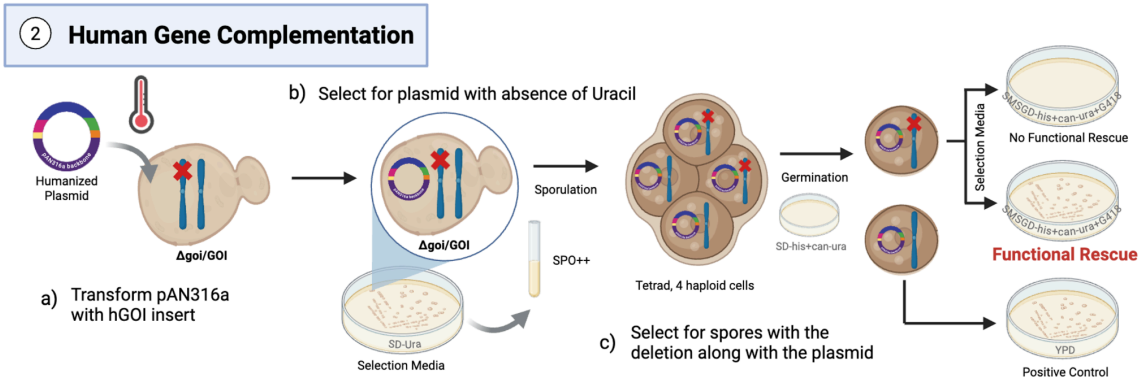


Figure 3. *Schematic workflow of human complementation, testing the ability of the human gene to rescue the function of its homologous yeast gene.*[4] *Created with BioRender.*

3) Human Gene Mutagenesis and Mutation-Pathogenicity Analysis

Tiled-Region Exchanged (T-REx) Mutagenesis developed by Kindree et al.[5] in the ANNB lab is a novel technique capable of simultaneously generating a pool of plasmids, each with a single missense mutation for each amino acid at each position along the gene of interest. This technique was employed on the human gene *TAMM41* within the TAM41 deletion strain.

3.1 Tiling Regions Within Human Gene of Interest

The cDNA sequence of the human gene of interest was input into a custom R-based pipeline to define optimal tiled regions for TREx mutagenesis. Tile design balanced the number and length of regions against oligonucleotide synthesis cost, as longer tiles require a greater number of NNK positions, each necessitating a separate oligonucleotide fragment within the pool. The final scheme partitioned the gene into mutagenizable segments while minimizing cost and maintaining experimental feasibility. The pipeline also generated primer pairs with BsaI overhangs flanking each tile.

3.2 Construction of ccdB-Containing Tile Plasmid Backbone

To enable high-efficiency replacement of mutagenized tiles, a ccdB counter-selection cassette was introduced into the humanized plasmid backbone in place of the tile of interest. The humanized yeast plasmid containing the human gene flanked by yeast promoter and terminator sequences (created in Section 2.4, Figure 2d) was amplified by inverse PCR to exclude the target tile region, generating a linearized backbone with BsaI overhangs. PCR products were verified by agarose gel electrophoresis and treated with DpnI to remove template plasmid. Amplified fragments were purified using magnetic bead-based purification. In parallel, the ccdB cassette with BsaI flanking overhangs was amplified and purified. The linearized backbone and ccdB fragment were assembled using Gibson Assembly to generate a plasmid, in which the tile of interest was replaced by the ccdB gene.

The assembled plasmids were transformed into ccdB-resistant *E. coli* strains for propagation following standard protocols as previously outlined. Individual colonies were isolated, and plasmid DNA was extracted by miniprep. Correct insertion of the ccdB cassette was verified by restriction digest using BsaI, which flanks the cassette, producing a diagnostic

fragment (~700 bp) corresponding to ccdB. Verified constructs were used as entry vectors for subsequent Golden Gate assembly.

3.3 Preparation of Oligonucleotide Libraries for T-REx Mutagenesis

Custom oligonucleotide pools (oPools) encoding mutagenized tiles were obtained as single-stranded DNA, briefly centrifuged to ensure complete recovery, and resuspended in 1/10 TE buffer to a final concentration of 10 µM. Each oligonucleotide contained a single degenerate NNK codon substitution at a defined position within the tiled region, enabling representation of all possible missense mutations.

To generate double-stranded DNA, oligo pools were subjected to a primer annealing and extension protocol. Each reaction (50 µL total volume) contained 27 µL autoclaved water, 5 µL of 10 µM oPool, 10 µL of 10 µM universal primer complementary to a constant flanking region present in all oligonucleotides (oAN22), and 5 µL of 10x NEBuffer 2.1. Samples were heated to 95 °C for 2 minutes and gradually cooled to room temperature to facilitate primer annealing using a thermocycler. Following annealing, 2.5 µL of 10 mM dNTPs and 0.5 µL of T4 DNA polymerase were added, mixed, and incubated under controlled conditions (5 min at 0 °C, 5 min at 22 °C, and 30 min at 37 °C) to enable strand extension.

Due to the strong exonuclease activity of T4 DNA polymerase, reactions were terminated promptly by the addition of 1 µL of 0.5 M EDTA to prevent over-digestion. The resulting double-stranded oligonucleotides were immediately purified using a column-based PCR purification kit to remove enzymes, salts, and residual single-stranded DNA. Purified oligonucleotide libraries were subsequently used for Golden Gate assembly to replace the ccdB cassette within the tiled region of the humanized plasmid.

3.4 Golden Gate Assembly of Mutagenized Tile Libraries

Golden Gate assembly was used to replace the ccdB cassette with mutagenized tiles from the oligonucleotide library. Reactions were assembled in a total volume of 20 µL using 1 µL of purified double-stranded oligonucleotide corresponding to the mutagenized tile, 4 µL of 5× NEBridge buffer, 1 µL of 20× Golden Gate enzyme mix (containing BsaI and ligase), and 500 ng of ccdB-containing plasmid backbone. The volume of plasmid backbone added was calculated based on its concentration to achieve 500 ng per reaction, and the remaining volume was adjusted with sterile water.

Reaction mixtures were thoroughly mixed and subjected to thermocycling conditions consisting of 60 cycles of digestion and ligation (1 min at 37 °C followed by 1 min at 16 °C), enabling iterative excision of the ccdB cassette and insertion of mutagenized tiles. Following cycling, reactions were incubated at 50 °C for 5 min to inactivate the restriction enzyme and then held at 4 °C until further use.

The assembled products were transformed into chemically competent *E. coli* DH5α cells for propagation. 2 µL of each Golden Gate plasmid pool product was added to the competent cells and transformed using a standard heat-shock protocol. Following recovery, 25 µL of transformed cells were diluted in 50 µL of sterile water and plated onto LB agar supplemented with ampicillin (25 mg/L) to assess transformation efficiency, with successful assembly and transformation indicated by high colony counts.

The remaining transformed culture (~475 µL) was transferred into 5 mL of ZYM505 medium supplemented with ampicillin (30 mg/L) and incubated at 30 °C overnight for amplification of the plasmid library, after which plasmid DNA was extracted using a standard miniprep protocol. An aliquot of this plasmid library was preserved by combining 800 µL of

overnight culture with 200 µL of cryoprotectant (50% glycerol, 35% DMSO) and stored at -80 °C. Purified plasmid DNA was quantified, and concentrations were adjusted to obtain ~1000 ng per sample for downstream yeast transformation.

3.5 Transformation of Variant Libraries into Yeast and Mutation Isolation

Purified plasmid libraries were subsequently introduced into yeast using a standard plasmid transformation protocol as described previously (Section 2.5, Figure 3). Following transformation, 10 µL of transformation (resuspended in 200 µL sterile water) was diluted in 50 µL and plated onto SD-ura to select for successful transformants. The remaining transformation mixture was inoculated into 5 mL of liquid SD–Ura and passaged twice to eliminate non-viable cells prior to downstream analysis. Cryoprotectant was added to the remaining pellet and stored in the -80 °C freezer for future experimentation.

After 48 hours of growth on SD–Ura plates, individual colonies were identified, with each colony assumed to harbor a distinct plasmid variant. 34 colonies were selected and each transferred into a 96-well plate containing 150 µL of SMSGD-ura+G418 per well. Two additional wells were inoculated with generated pathogenic control variants derived from site-directed mutagenesis. Cultures were incubated at 30 °C for 24 hours.

3.6 Sporulation and Germination Assay for Functional Assessment

Following growth in well plates, cultures were arrayed in a 6 × 6 grid onto SPO++ agar plates to induce sporulation. After incubation under sporulation conditions (5 days at room temperature), cells were replica plated onto SD+can-his-ura germination agar and incubated at 30 °C for 2 days to select for haploid Mat α spores carrying the plasmid.

Cells were then sequentially replica plated onto a YPD plate and two SMSGD-his+can-ura+G418 selection plates. One selection plate was incubated at 37 °C to

approximate physiological conditions, while the other selection plate along with the YPD positive control were incubated at 30 °C for optimal *S. cerevisiae* growth.

Cell growth was assessed at 24 and 48 hours post-plating. Growth on selective media indicated successful complementation of the essential yeast gene by the human variant, whereas absence of growth indicated loss of function. Thus, the assay provided a binary readout of variant pathogenicity (viable vs. non-viable). See Figure 4.

3.7 Sequence Verification of Yeast Variants by Colony PCR

To confirm plasmid variant identity, colony PCR was performed following enzymatic lysis. Yeast colonies were picked from germination plates (prior to replica plating) and treated with lysis buffer containing zymolyase and dithiothreitol (DTT) to digest the cell wall and release DNA. Samples were incubated at 37 °C for 1 hour, followed by heat inactivation.

The resulting lysates were used directly as PCR templates to amplify the inserted gene within pAN316a using flanking SP6 and T7 primers. PCR products were preliminarily verified by agarose gel electrophoresis, purified via magnetic bead-based purification, and submitted for sequencing to identify the specific mutation present in each variant assayed.

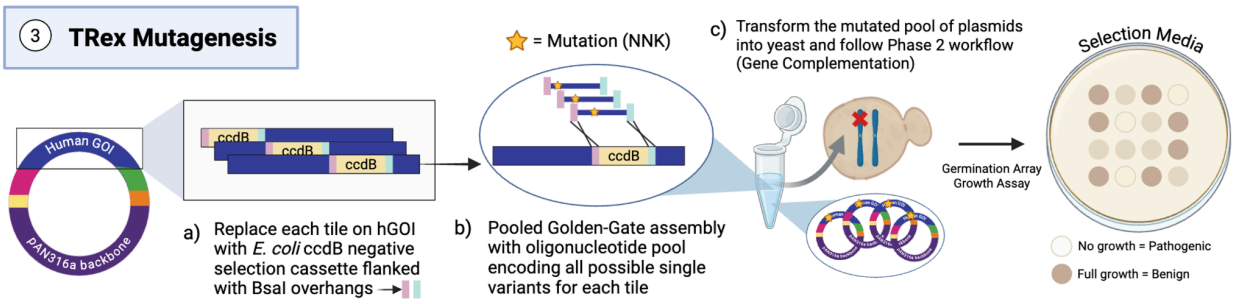


Figure 4*. T-REx mutagenesis workflow. Mutagenized oligonucleotides are inserted to their corresponding tile, replacing a ccdB cassette via Golden Gate assembly to generate variant plasmid libraries. The plasmid libraries are propagated in E. coli and the purified plasmid pool*

is subsequently introduced into yeast. Functional complementation is assessed as a pathogenicity metric using a germination assay with binary growth readout.

Results

Functional Complementation of Yeast Deletion Strains

Functional complementation was assessed by testing the ability of the human homologues to rescue growth of corresponding yeast deletion strains under gene-specific selective conditions (Table 3). In reference to Figure 5, growth on selective media indicates that the introduced gene can functionally replace the deleted endogenous gene, whereas absence of growth indicates failure to complement. YPD plates served as non-selective controls for cell viability.

Robust growth was observed for all strains on YPD plates, confirming successful sporulation, germination, and overall cell viability across all conditions.

Under respective selective conditions, SPE3, MPC2, VMS1, and PRP4 deletion strains grew only when transformed with their respective yeast genes; no growth occurred with the empty vector or human constructs. This lack of functional complementation suggests that these human homologs are not sufficiently conserved to substitute for their yeast counterparts under the tested conditions, precluding their use in this yeast-based variant study.

For RLP24, no growth was observed on selective media for any construct, including the yeast gene control, despite robust growth on YPD. While the human gene did not restore growth, the absence of complementation by the yeast gene indicates that the assay was not successful for this strain, and no conclusions can be drawn without further testing.

TAM41 deletion strains exhibited growth on selective media when transformed with both the yeast gene or the human gene, while no growth was observed with the empty vector, further showing its essentiality. This successful complementation confirms that human TAMM41 can compensate for the essential function of yeast TAM41, providing a valid platform for downstream pathogenicity analysis of missense mutations. See Figure 5 below.

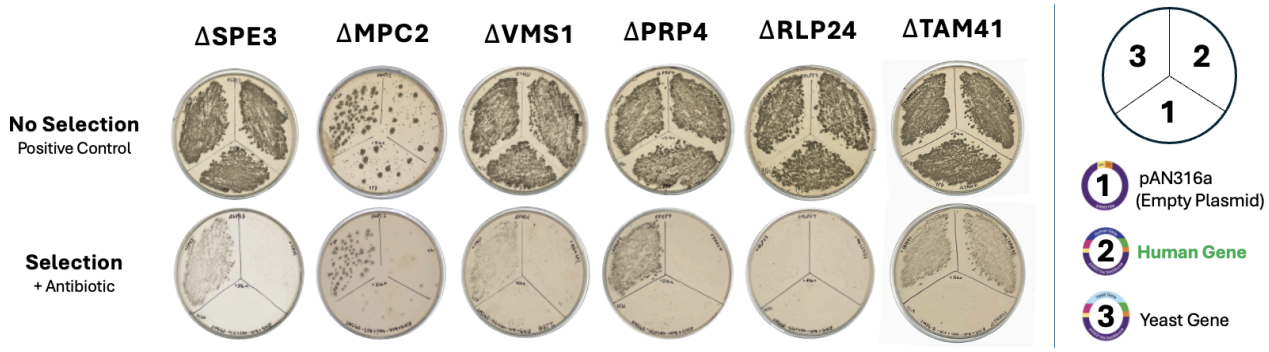


Figure 5. *Functional complementation of yeast deletion strains by yeast and human orthologs. Yeast deletion strains for SPE3, MPC2, VMS1, PRP4, RLP24, and TAM41 were transformed with empty vector (pAN316a), yeast gene (positive control), and human homolog plasmids. Following sporulation and germination, haploid spores were selected and replica plated onto YPD (non-selective control) and antibiotic selective media to isolate growth of deletion strains containing a plasmid. YPD plates confirmed viability of all strains. Growth on selective media indicates functional complementation of the deleted gene by the plasmid-borne construct. Only TAM41 showed complementation by both yeast and human genes, while SPE3, MPC2, VMS1, and PRP4 were complemented exclusively by yeast genes. RLP24 showed no growth under selective conditions for any construct.*

Functional Assessment of TAMM41 Variants

To systematically evaluate the functional impact of genetic variation within TAMM41, a total of 36 missense variants were selected for qualitative growth assays. Assays were performed on Mat α haploid spores isolated using the SD+can-his-ura selection system to ensure the growth observed was solely due to the human variant. Variants 1-34 were generated via T-REx mutagenesis on a single tile region of amino acids from position 40 to 87 on TAMM41.

The variants were assessed for their ability to support growth on selective germination media at 30°C (optimal yeast temperature) and 37°C (simulated physiological temperature), with growth recorded at 24 and 48 hours. As seen in Figure 6, all variants showed robust growth on YPD positive control plates at both time marks, confirming the viability of the assayed spores.

At 30°C, 9 variants (numbered 2, 7, 9, 10, 17, 18, 23, 26, and 30) failed to support growth on selective media, indicating a complete loss-of-function under standard conditions. These 9 variants similarly failed to grow at 37°C. Furthermore, an additional 5 variants (6, 13, 15, 32, and 35) exhibited temperature-sensitive phenotypes; these variants supported growth at 30°C but failed to grow at 37°C on selection media.

Known pathogenic controls were included for assay validation. Variant 35 (Y110C) showed growth at 30 °C but not at 37 °C, displaying temperature-sensitive loss of function, whereas variant 36 (P137L) retained functional growth under both temperature conditions. These results provide a binary (viable vs. non-viable) qualitative map of variant pathogenicity, identifying both strong loss-of-function and conditionally destabilizing mutations within the TAMM41 sequence (Figure 6, Appendix A: Table 4)

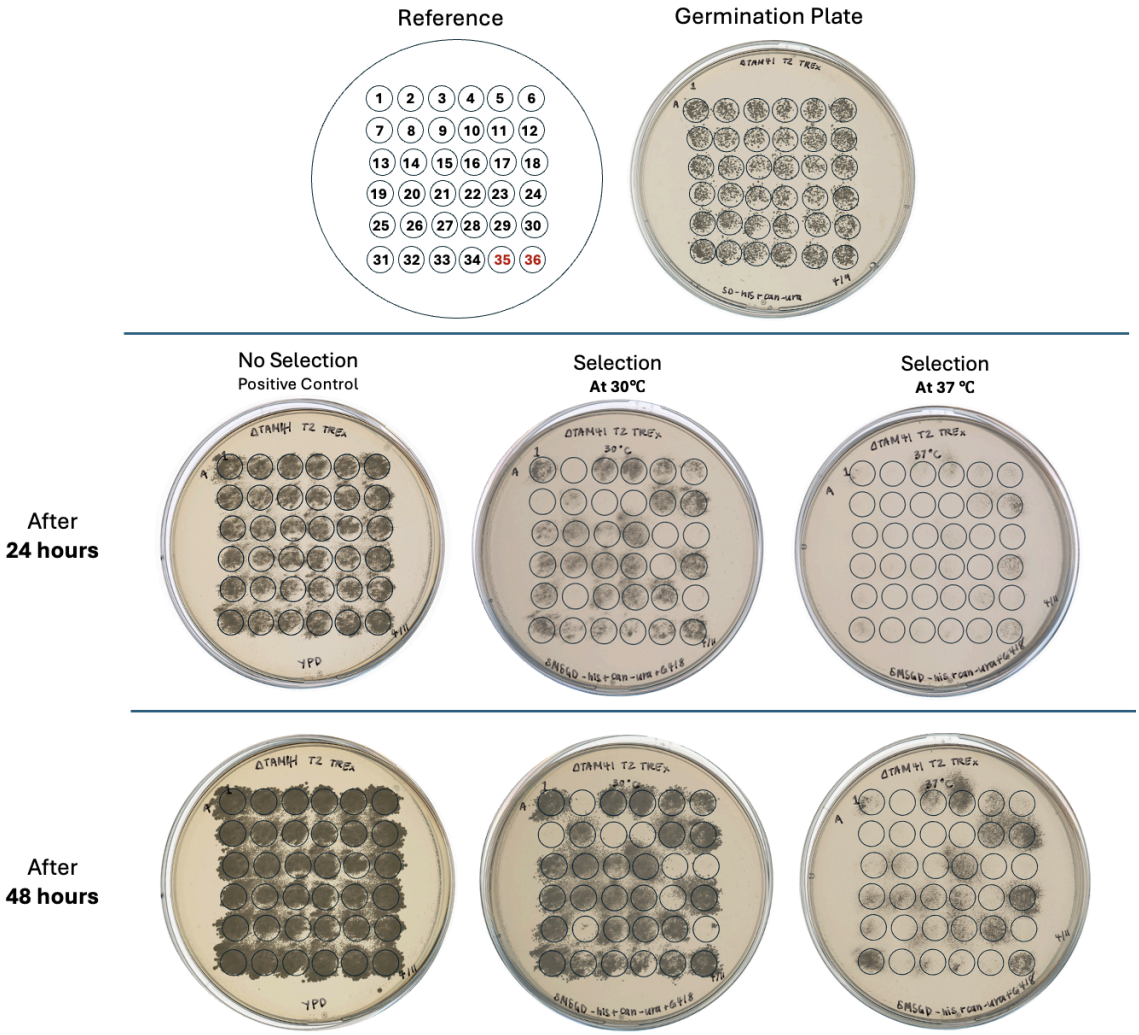


Figure 6. *Functional assessment of 36 TAMM41 missense variants.*

Representative growth arrays for TAMM41 variants are shown on YPD (control) and selective germination media at 30°C and 37°C, captured after 24 and 48 hours of incubation. All variants exhibited robust growth on YPD. Nine variants (2, 7, 9, 10, 17, 18, 23, 26, and 30) displayed no growth at 30°C, indicating a loss of essential protein function. Five additional variants (6, 13, 15, 32, and 35) showed temperature-sensitive defects, with growth at 30°C but not at 37°C. Two known pathogenic variants were generated; Variant 35 (Y110C) demonstrated temperature-dependent growth failure, while variant 36 (P137L) remained functional across both conditions. Compare with Appendix A: Table 4 for variant sequence data.

Discussion

The results of this study demonstrate the utility and limitations of using a humanized *Saccharomyces cerevisiae* platform to map the functional impact of human genetic variation. By utilizing the yeast life cycle and essential gene lethality, we successfully established a diagnostic framework for the mitochondrial gene TAMM41, while identifying critical biological barriers to functional complementation in other disease-relevant genes.

Functional Complementation Assay Analysis

Functional complementation in the humanized yeast system is dependent on both conservation of core protein function and compatibility with the cellular environment. While sequence homology suggests potential functional similarity, successful rescue additionally requires proper protein folding, localization, and interaction with species-specific cofactors and complexes. As a result, lack of complementation does not necessarily indicate absence of conserved function, but rather highlights limitations in cross-species compatibility. To better understand these outcomes, each gene was evaluated individually in the context of its biological role and assay conditions.

SPE3, involved in polyamine biosynthesis, was assessed under pantothenic acid limited conditions, where pathway flux becomes critical for growth.[7] While the yeast gene restored viability, the human homolog did not. This may reflect differences in cofactor usage or regulation of polyamine synthesis between yeast and humans, particularly under nutrient-limited conditions.[8] Additionally, metabolic pathway integration is often tightly regulated, and incomplete compatibility with upstream or downstream enzymes may impair function of the human protein in yeast.

MPC2 functions as part of the mitochondrial pyruvate carrier complex and was assessed under leucine-deficient conditions, which increase reliance on mitochondrial metabolism.[18] The inability of the human homolog to rescue may be attributed to the requirement for proper assembly with other complex subunits, where incomplete conservation of interacting partners could impair transport activity despite structural similarity. In addition to these biological factors, aspects of the experimental design, like growth under leucine auxotrophic conditions, imposes additional metabolic constraints which may reduce overall recovery of viable colonies. Consistent with this, low germination efficiency was observed on SD-leu+can-ura germination plates and reflected on subsequent replica plates (Figure 5). Furthermore, MPC2 deletion strains frequently exhibited slow growth, indicating an inherent fitness defect under these conditions. Together, these factors suggest that both biological incompatibility and condition-specific experimental constraints contributed to the lack of observed complementation, and that retesting under optimized conditions would be required for definitive interpretation.

VMS1 plays a role in mitochondrial protein quality control and was tested under heat stress at 37 °C where this gene becomes essential. The lack of complementation by the human homolog may reflect differences in stress response pathways or the requirement for species-specific cofactors involved in ribosome-associated quality control. Stress-induced phenotypes are particularly sensitive to subtle differences in protein function, which may not be tolerated in this context.[7,12]

PRP4 is a serine/threonine kinase involved in spliceosome activation, a highly complex and tightly regulated process. The absence of complementation by the human homolog is likely due to evolutionary divergence in RNA processing machinery, which may prevent proper integration of the human protein within the yeast spliceosomal system. Differences in kinase

substrate specificity or spliceosomal architecture may impair the ability of the human kinase to recognize and phosphorylate yeast targets. Given the extensive protein–protein interactions required for spliceosome assembly and activation, even minor incompatibilities are likely sufficient to prevent successful functional substitution.[7,8,13]

RLP24, an essential factor in ribosome biogenesis, did not exhibit growth for any construct, including the yeast control, suggesting a technical or strain-specific limitation rather than a true biological outcome. While factors such as inefficient sporulation, or insufficient gene expression may contribute, variability in the deletion strain itself is also a likely source of error. Inconsistent essentiality observed across independently derived deletion strains indicates that background genetic differences or suppressor mutations may influence viability, even when the same gene is disrupted. As ribosome assembly is required for survival, successful rescue by the yeast gene would be expected under appropriate conditions. Repeating the assay with independent strains, alongside verification of plasmid integrity and expression, are necessary to resolve this discrepancy.

TAM41 was the only gene to demonstrate successful complementation by its human homolog, enabling downstream variant analysis. TAM41 functions in cardiolipin biosynthesis within the mitochondrial inner membrane, a pathway that appears to be sufficiently conserved to support cross-species functionality. This suggests that enzymes involved in core metabolic processes with fewer interacting partners may be more amenable to functional substitution. The successful complementation of TAMM41 validates the experimental framework and supports its use for variant effect mapping in compatible gene systems.

Together, these findings demonstrate that functional complementation in yeast is highly gene-dependent and influenced by both evolutionary conservation and cellular context. In some

cases (such as MPC2 and VMS1), human proteins may lack mitochondrial targeting sequences compatible with the yeast import machinery, resulting in mislocalization to the cytosol rather than the mitochondria. Additionally, human cDNA may not be expressed at sufficient levels in yeast or may require post-translational modifications that are not supported in this system. These limitations highlight key factors influencing cross-species compatibility.[19] Future work is aimed at optimizing expression, localization, and assay conditions, as well as expanding the range of compatible genes, will further enhance the utility of this approach.

Functional Landscape of TAMM41 Variants and Clinical Implications

Functional assessment of 36 missense variants within amino acids 40-87 (deemed Tile 2) of TAMM41 revealed a range of phenotypes, including complete loss-of-function, temperature-sensitive effects, and wild-type-like behavior. Variants that failed to support growth at both 30 °C and 37 °C—D50Q, G80S, G80C, K77V, N69K, and G80R—were classified as loss-of-function, indicating severe disruption of protein activity. In contrast, variants such as P58L, A47F, M48N, N65S, and K64P supported growth at 30 °C but not at 37 °C, consistent with temperature-sensitive phenotypes. These mutations likely impair protein stability or folding, where function is retained under permissive conditions but lost under thermal stress, demonstrating that standard growth conditions alone may underestimate functional defects. See Appendix A: Table 4. The ability to resolve these distinct phenotypic classes highlights the utility of this platform in converting genetic variation into measurable functional outcomes.

With regards to the known pathogenic controls included to support the validity of the assay, Y110C (Variant 35) exhibited a strong loss-of-function phenotype as expected, while P137L (Variant 36) showed no detectable growth defect. This discrepancy highlights an inherent

limitation of the system: yeast-based assays are highly sensitive to disruptions in essential cellular processes and do not fully recapitulate human biology. As a result, variants that appear functionally neutral in yeast may still be pathogenic in humans, while some detected functional impairments may not reach the threshold required to cause disease.

The reliability of this platform can be contextualized using a score for concordance between yeast-based functional outcomes and known human phenotypes, which is relatively high but not absolute.[20] Consequently, yeast-derived results should not be used as the sole basis for clinical decision-making, but rather as a preliminary tool for variant interpretation that can inform further investigation, particularly in cases where patients present with relevant symptoms.

Several variants were excluded from analysis due to sequencing limitations or the presence of secondary mutations, including large deletions and low-quality reads, underscoring the importance of sequence validation in high-throughput workflows.

The relevance of these findings is underscored by the role of TAMM41 in mitochondrial function, encoding mitochondrial phosphatidate cytidylyltransferase required for cardiolipin biosynthesis which is a critical component of mitochondrial membrane integrity and oxidative phosphorylation. Disruption of this function is associated with Combined Oxidative Phosphorylation Deficiency (COXPD), a rare mitochondrial disorder characterized by impaired energy metabolism among other symptoms.[15] The identification of loss-of-function and conditionally destabilizing variants therefore provides direct insight into mutations that may compromise mitochondrial function and contribute to disease.

Overall, these results demonstrate that yeast complementation with T-REx mutagenesis enables efficient discrimination between deleterious, conditionally impaired, and tolerated variants. This approach provides a scalable framework for constructing variant effect maps,

while also defining the boundaries of its applicability and emphasizing the need for integration with complementary methods in the interpretation of human genetic variation.

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Appendix A

Table 4. *Summary of missense variants generated within Tile 2 (amino acids 40-87) of TAMM41 and their corresponding growth phenotypes at 30 °C and 37 °C. Mutations were introduced using TREx mutagenesis, with two known pathogenic variants included via site-directed mutagenesis. Growth was assessed following functional complementation in yeast, where "x" indicates no growth. Variants were classified based on their ability to support viability at standard (30 °C) and elevated (37 °C) temperatures. All variants were sequenced with Oxford Nanopore PromethION, and additional notes included where necessary.*

Name	Mutation	Codon	Growth @ 30°C	Growth @ 37°C	Additional Notes
1	K82L	AAA -> TTG			
2	D50Q	GAT -> CAG	x	x	
3	M48W	AGT -> TGG			
4	I83R	ATA -> AGG			
5	T54P	ATC -> CCT			
6	P58L	CCA -> CGT		x	
7	G80S	GGC -> AGC	x	x	
8	A47F	GCT -> TTT		x	
9	G80C	GCC -> TGT	x	x	Additional large deletion from 40-75
10			x	x	Inconclusive (Low Quality)
11	A47T	GCT -> ACG			
12	K68W	AAG -> TGG			
13	M48N	ATG -> AAT		x	
14	S86V	TCT -> GTT			
15	N65S	ACC -> TCG		x	
16	V55C	GTG -> TGT			
17	Y73W	TAC -> TGG	x	x	Low Quality Read

18			x	x	Inconclusive (Low Quality)
19	K82S	AAA -> TCG			
20	S63M	TCT -> AGT			
21	W70Q	TCG -> CAG			
22	K68W	AAG -> TGG			
23	K77V	AAA -> GTG	x	x	
24	L79F	TTA -> TTT			
25	M48G	ATG -> GGG			
26	N69K	AAC -> AAG	x	x	*39-54 deletion on forward read
27	V59A	GTC -> GCT			
28	V78Q	GTG -> CAG			
29	V78A	GTG -> TCT			
30	G80R	GCC -> CGG	x	x	
31	W61F	TGG -> TTT			
32	K64P	AAG -> CCG		x	
33	P58P	CCA -> CCT			
34	K64V	AAG -> GTG			
35	Y110C	TAT -> TGT		x	Known pathogenic control (Generated via SDM)
36	P137L	CCA -> CTA			Known pathogenic control (Generated via SDM)